

MAT1002 Lecture 9,

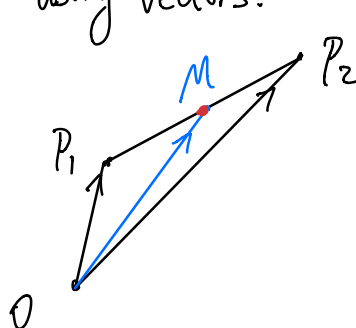
Outline

- Vectors
- Dot products
- Cross products

Vectors

Midpoint Formula

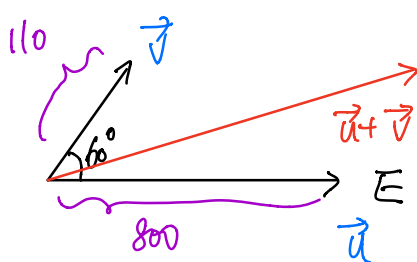
If midpoint of the line segment P_1P_2 joining $P_1(x_1, y_1, z_1)$ & $P_2(x_2, y_2, z_2)$ is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2})$. This can be proven using vectors:


$$\begin{aligned}\vec{OM} &= \vec{OP_1} + \frac{1}{2} \vec{P_1P_2} \\ &= \langle x_1 + \frac{x_2 - x_1}{2}, y_1 + \frac{y_2 - y_1}{2}, z_1 + \frac{z_2 - z_1}{2} \rangle \\ &= \langle \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \rangle.\end{aligned}$$

$$\text{Hence } M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}).$$

Applications of Vectors

EXAMPLE 8 A jet airliner, flying due east at 800 km/h in still air, encounters a 110 km/h tailwind blowing in the direction 60° north of east. The airplane holds its compass heading due east but, because of the wind, acquires a new ground speed and direction. What are they?



Sol: $\vec{u} = \langle 800, 0 \rangle$.

$$v_1 = 110 \cdot \cos \frac{\pi}{3} = 55$$

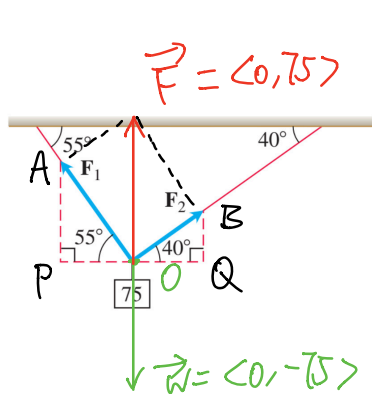
$$v_2 = 110 \cdot \sin \frac{\pi}{3} = 55\sqrt{3}$$

$$\vec{u} + \vec{v} = \langle 855, 55\sqrt{3} \rangle$$

$$\text{Speed} = S = |\vec{u} + \vec{v}| = \sqrt{855^2 + 55^2 \cdot 3} \quad (\approx 860.3) \text{ km/h}$$

$$\text{Direction} = \frac{\vec{u} + \vec{v}}{|\vec{u} + \vec{v}|}$$

EXAMPLE 9 A 75-N weight is suspended by two wires, as shown in Figure 12.18a. Find the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ acting in both wires.



Sol : $\cdot \mathbf{F}_1 + \mathbf{F}_2 = \mathbf{F} = \langle 0, 75 \rangle$.

$\cdot \mathbf{F}_1 = \langle -|\mathbf{F}_1| \cos 55^\circ, |\mathbf{F}_1| \sin 55^\circ \rangle$ (*)

$\mathbf{F}_2 = \langle |\mathbf{F}_2| \cos 40^\circ, |\mathbf{F}_2| \sin 40^\circ \rangle$

$\cdot \begin{cases} 0 = |\mathbf{F}_2| \cos 40^\circ - |\mathbf{F}_1| \cos 55^\circ \\ 75 = |\mathbf{F}_2| \sin 40^\circ + |\mathbf{F}_1| \sin 55^\circ \end{cases}$

$\Rightarrow$ Solve for $|\mathbf{F}_1|$ and $|\mathbf{F}_2|$: $|\mathbf{F}_1| \approx 57.67 \text{ (N)}$
and $|\mathbf{F}_2| \approx 43.18 \text{ (N)}$.

Plug back to (*) to get $\mathbf{F}_1$ and $\mathbf{F}_2$.

Dot Products

Definition

Given two vectors $\vec{u} = \langle u_1, u_2, \dots, u_n \rangle$ and $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$, the **dot product** of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \cdot \vec{v}$, is defined by

$$\vec{u} \cdot \vec{v} := u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

Example

(a) $\langle 1, -2, -1 \rangle \cdot \langle -6, 2, -3 \rangle = (1)(-6) + (-2)(2) + (-1)(-3)$
 $= -6 - 4 + 3 = -7$

(b) $\left(\frac{1}{2} \mathbf{i} + 3\mathbf{j} + \mathbf{k} \right) \cdot (4\mathbf{i} - \mathbf{j} + 2\mathbf{k}) = \left(\frac{1}{2} \right)(4) + (3)(-1) + (1)(2) = 1$

Properties of the Dot Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and c is a scalar, then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$

3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$

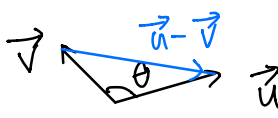
5. $\mathbf{0} \cdot \mathbf{u} = 0$.

Proofs are easy, e.g.

4. $\vec{u} \cdot \vec{u} = u_1^2 + u_2^2 + \dots + u_n^2 = \left(\sqrt{u_1^2 + u_2^2 + \dots + u_n^2} \right)^2 = |\vec{u}|^2$.

Dot Products & Angles in $\mathbb{R}^2$ & $\mathbb{R}^3$

Consider $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^2$ (or $\mathbb{R}^3$), with angle θ made between them ($0 \leq \theta \leq \pi$). Then by the law of cosine,


$$\begin{aligned} |\vec{u} - \vec{v}|^2 &= |\vec{u}|^2 + |\vec{v}|^2 - 2\cos\theta |\vec{u}| |\vec{v}| \\ (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) \end{aligned}$$

$$\Rightarrow \vec{u} \cdot \vec{u} - \vec{u} \cdot \vec{v} - \vec{v} \cdot \vec{u} + \vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} - 2\cos\theta |\vec{u}| |\vec{v}|$$

$$\Rightarrow$$

$$\vec{u} \cdot \vec{v} = \cos\theta |\vec{u}| |\vec{v}|.$$

Remarks

• If $\vec{u}$ and $\vec{v}$ are nonzero, then

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \Rightarrow \theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}\right).$$

• If $\vec{u}$ and $\vec{v}$ are unit vectors, then

$$\cos \theta = \vec{u} \cdot \vec{v} \Rightarrow \theta = \arccos(\vec{u} \cdot \vec{v})$$

$$\cdot \vec{u} \cdot \vec{v} = 0 \Leftrightarrow (\vec{u} = \vec{0} \text{ or } \vec{v} = \vec{0} \text{ or } \theta = \frac{\pi}{2}).$$

Generalizing from $\mathbb{R}^2$ and $\mathbb{R}^3$:

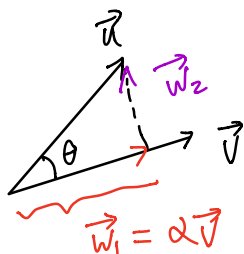
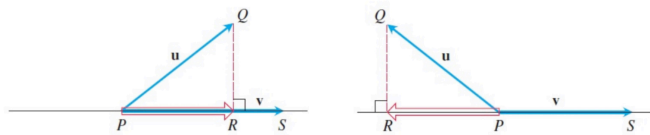
Def: For $\vec{u}, \vec{v}$ in $\mathbb{R}^n$:

• If $\vec{u} \neq \vec{0}$ and $\vec{v} \neq \vec{0}$, then the **angle made by $\vec{u}$ and $\vec{v}$** is $\theta := \arccos\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}\right)$.

• $\vec{u}$ and $\vec{v}$ are **orthogonal** (or **perpendicular**) if $\vec{u} \cdot \vec{v} = 0$ (note that $\vec{u}$ and $\vec{v}$ can be zero).

Vector Projections

Consider the following figure. Geometrically, the vector projection of $\vec{u} = \overrightarrow{PQ}$ onto a nonzero vector $\vec{v} = \overrightarrow{PS}$, denoted by $\text{proj}_{\vec{v}} \vec{u}$, is the vector $\overrightarrow{PR}$ determined by dropping a perpendicular from Q to the line PS .



$$\vec{u} = \alpha \vec{v} + \vec{w}_2 ; \vec{w}_2 \perp \vec{v}$$

$$\Rightarrow \vec{u} \cdot \vec{v} = \alpha \vec{v} \cdot \vec{v} + \cancel{\vec{w}_2 \cdot \vec{v}} = \alpha |\vec{v}|^2$$

$$\Rightarrow \alpha = (\vec{u} \cdot \vec{v}) / |\vec{v}|^2.$$

Definition: If $\vec{v} \neq \vec{0}$ and θ is the angle between $\vec{u}$ and $\vec{v}$, then the **vector projection** of $\vec{u}$ onto $\vec{v}$ is defined by

$$\text{proj}_{\vec{v}} \vec{u} := \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \right) \frac{\vec{v}}{|\vec{v}|} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}.$$

The number $(\vec{u} \cdot \vec{v})/|\vec{v}|$ is called the **scalar component** of $\vec{u}$ in the direction of $\vec{v}$.

$$\text{proj}_{\vec{v}} \vec{u} = \underbrace{\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \right)}_{\text{Scalar component}} \underbrace{\left(\frac{\vec{v}}{|\vec{v}|} \right)}_{\text{Direction of } \vec{v}}.$$

Scalar component = $(\cos \theta) |\vec{u}|$

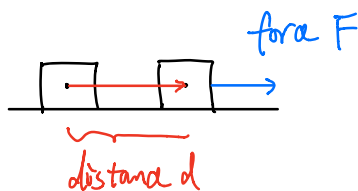
EXAMPLE Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$.

Sol: $\vec{u} \cdot \vec{v} = 6 - 3 \cdot 2 - 2 \cdot 2 = -4$; $|\vec{v}| = 3$

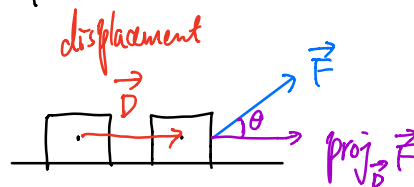
$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} = \frac{-4}{9} \langle 1, -2, -2 \rangle.$$

$$\text{Scalar component} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} = -\frac{4}{3}.$$

e.g. Work done by constant force $\vec{F}$.



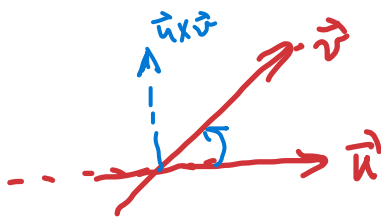
$$W = Fd.$$



$$W = |\text{proj}_{\vec{B}} \vec{F}| |\vec{B}| = \frac{\vec{F} \cdot \vec{B}}{|\vec{B}|} \cdot |\vec{B}| = \vec{F} \cdot \vec{B}$$

$$W = \vec{F} \cdot \vec{B}.$$

Cross Products



$$\vec{u} = \lambda \vec{v} \quad \lambda = 2$$

$$\vec{u} = 2\vec{v}$$

$$\vec{u} = \lambda \vec{v}$$

$$|\vec{u}| = |\lambda| |\vec{v}|$$

$$|\lambda| = \frac{|\vec{u}|}{|\vec{v}|}$$

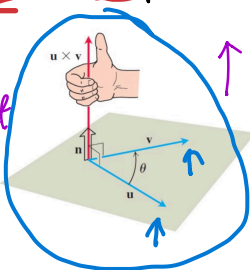
Two nonzero vectors $\vec{u}$ and $\vec{v}$ are said to be parallel if $\vec{u} = \lambda \vec{v}$ for some constant λ . If $\vec{u}$ and $\vec{v}$ are nonzero and not parallel, then they determine a plane $P_{\vec{u}, \vec{v}}$.

Given two nonparallel, nonzero vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^3$, the cross product $\vec{u} \times \vec{v}$ is a vector that satisfies:

• $(\vec{u} \times \vec{v}) \perp P_{\vec{u}, \vec{v}}$, and;

• The triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ satisfies the "right-hand rule", as follows:

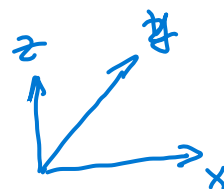
Positive angle measurement from $\vec{u}$ to $\vec{v}$



$\vec{u} \times \vec{v}$ direction

$$\vec{u} \times \vec{v} \perp \vec{u}$$

$$\vec{u} \times \vec{v} \perp \vec{v}$$



Definition

Given vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ in $\mathbb{R}^3$, the cross product of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \times \vec{v}$, is defined by

$$\vec{u} \times \vec{v} := \langle u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1 \rangle$$

$$\vec{w} = \langle w_1, w_2, w_3 \rangle$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Remark: It is defined even if $\vec{u}$ and $\vec{v}$ are parallel or $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$.

Determinants

For each $n \times n$ matrix

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, \text{ there is a}$$

$$A = \begin{pmatrix} v_1 & v_2 \\ u_1 & u_2 \end{pmatrix}$$

$$\vec{v} = \langle v_1, v_2 \rangle$$

Special number $\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$ called the determinant.

2x2 and 3x3 determinants can be computed as follows

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} := \underline{ad - bc}.$$

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} := a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

(+) (-) (+)

A note on determinants will be posted separately.

In determinant notation, $\vec{u} \times \vec{v}$ can be memorized as

$$\vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Example

If $\vec{u} = \langle 2, 1, 1 \rangle$ and $\vec{v} = \langle -4, 3, 1 \rangle$, then

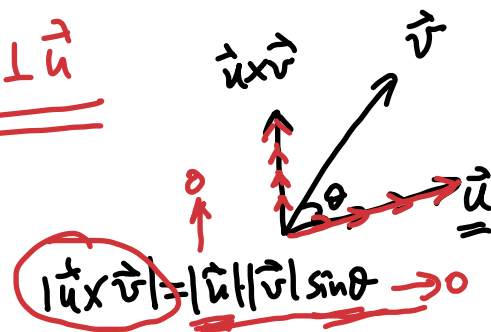
$$\begin{aligned} \vec{u} \times \vec{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \mathbf{k} \\ &= -2\mathbf{i} - 6\mathbf{j} + 10\mathbf{k} \\ &= \langle -2, -6, 10 \rangle \end{aligned}$$

$$\begin{aligned} &= (u_2 v_3 - u_3 v_2) \vec{i} \\ &\quad - (u_1 v_3 - v_1 u_3) \vec{j} \\ &\quad + (u_1 v_2 - v_1 u_2) \vec{k} \end{aligned}$$

Remark

It is straightforward to check that $(\vec{u} \times \vec{v}) \cdot \vec{u} = (\vec{u} \times \vec{v}) \cdot \vec{v} = 0$, so $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$.

$$\underline{\underline{\vec{u} \times \vec{v} \perp \vec{u}}}$$



$$\vec{u} \times \vec{v} = \left\langle \frac{u_2 v_3 - u_3 v_2}{u_2}, \frac{u_3 v_1 - u_1 v_3}{u_2}, \frac{u_1 v_2 - u_2 v_1}{u_3} \right\rangle$$

$$\vec{u} = \langle \underline{u_1}, \underline{u_2}, \underline{u_3} \rangle$$

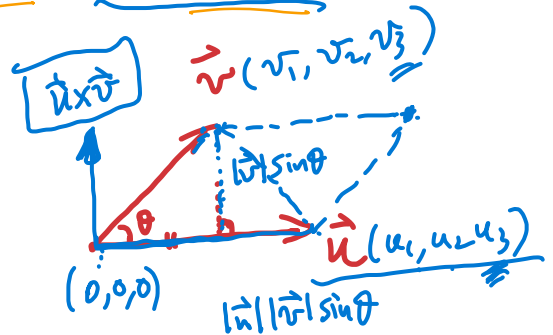
Check:

$$(\vec{u} \times \vec{v}) \cdot \vec{u} = \underline{u_2 v_3 - u_3 v_2} (\underline{u_1}) + (\underline{u_3 v_1 - u_1 v_3}) (\underline{u_2}) + (\underline{u_1 v_2 - u_2 v_1}) (\underline{u_3})$$

$$= 0$$

$(\vec{u} \times \vec{v}) \cdot \vec{v} = 0$ is similar.

Length of Cross Products



Theorem

Let $\vec{u}$ and $\vec{v}$ be vectors in $\mathbb{R}^3$, and let θ be the angle between them (where $0 \leq \theta \leq \pi$). Then

$$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta.$$

area of the parallelogram formed by $\vec{u}$ and $\vec{v}$

$$\begin{aligned} \text{Proof: } |\vec{u}|^2 |\vec{v}|^2 \sin^2 \theta &= |\vec{u}|^2 |\vec{v}|^2 (1 - \cos^2 \theta) \\ &= |\vec{u}|^2 |\vec{v}|^2 - (|\vec{u}| |\vec{v}| \cos \theta)^2 = |\vec{u}|^2 |\vec{v}|^2 - (\vec{u} \cdot \vec{v})^2 \\ &= (u_1^2 + u_2^2 + u_3^2)(v_1^2 + v_2^2 + v_3^2) - (u_1 v_1 + u_2 v_2 + u_3 v_3)^2 \\ &= u_2^2 v_3^2 - 2 u_2 u_3 v_2 v_3 + u_3^2 v_2^2 + u_3^2 v_1^2 - 2 u_1 u_3 v_1 v_3 + u_1^2 v_3^2 \\ &\quad + u_1^2 v_2^2 - 2 u_1 u_2 v_1 v_2 + u_2^2 v_1^2 \\ &= (u_2 v_3 - u_3 v_2)^2 + (u_3 v_1 - u_1 v_3)^2 + (u_1 v_2 - u_2 v_1)^2 \\ &= |\vec{u} \times \vec{v}|^2 \end{aligned}$$

$$\vec{u} \times \vec{v}$$

$$= \langle w_1, w_2, w_3 \rangle$$

$$|\vec{u} \times \vec{v}| = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

$$|\vec{u} \times \vec{v}|^2 = w_1^2 + w_2^2 + w_3^2$$

$$\Rightarrow |\vec{u}| |\vec{v}| \sin \theta = |\vec{u} \times \vec{v}| \quad (\text{since } \sin \theta \geq 0).$$



Corollary (= consequence of thm)

For any vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^3$,

$$\vec{u} \times \vec{v} = \vec{0}$$

if and only if $\vec{u} = \vec{0}$, or $\vec{v} = \vec{0}$, or $\vec{u}$ and $\vec{v}$ are parallel.

$$\vec{u} \cdot \vec{v} = 0 \\ \Rightarrow \vec{u} \perp \vec{v}$$

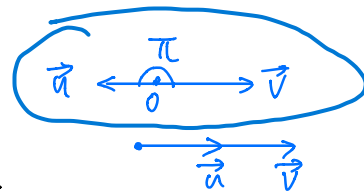
$$\vec{u} \times \vec{v} = \vec{0} \\ \Rightarrow \vec{u} \parallel \vec{v}$$

$$|\vec{u} \times \vec{v}| = \underbrace{|\vec{u}|}_{\uparrow} \underbrace{|\vec{v}|}_{\uparrow} \underbrace{\sin \theta}_{\uparrow} = 0$$

Proof: $\vec{u} \times \vec{v} = \vec{0} \Leftrightarrow |\vec{u} \times \vec{v}| = 0$

$$\Leftrightarrow |\vec{u}| = 0, |\vec{v}| = 0, \text{ or } \boxed{\sin \theta = 0}$$

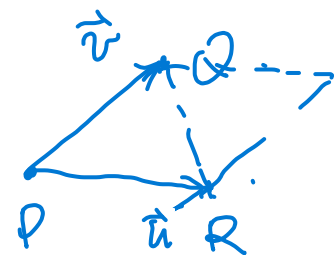
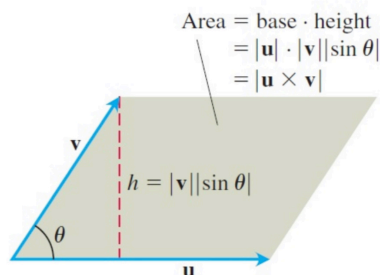
$$\Leftrightarrow \vec{u} = \vec{0}, \vec{v} = \vec{0}, \theta = 0 \text{ or } \theta = \pi.$$



$$\theta = 0 \\ \theta = \pi.$$



Since $|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}|\sin \theta$, where θ is the angle between $\vec{u}$ and $\vec{v}$, the quantity $|\vec{u} \times \vec{v}|$ is equal to the area of the parallelogram determined by $\vec{u}$ and $\vec{v}$, as shown in the following figure.



Example

Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$ and $R(-1, 1, 2)$.

Solution

The area is $3\sqrt{2}$.

$$\vec{u} = \vec{PR} = \langle -2, 2, 2 \rangle$$

$$\vec{v} = \vec{PQ} = \langle 1, 2, -1 \rangle$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & 2 \\ 1 & 2 & -1 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 2 \\ 2 & -1 \end{vmatrix} - \hat{j} \begin{vmatrix} -2 & 2 \\ 1 & -1 \end{vmatrix} + \hat{k} \begin{vmatrix} -2 & 2 \\ 1 & 2 \end{vmatrix} \\ = -6\hat{i} - 0\hat{j} + (-6)\hat{k}$$

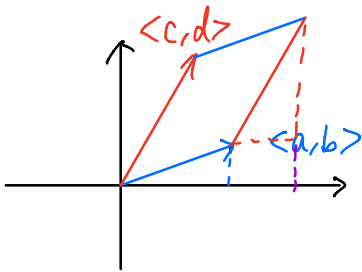
$$|\vec{u} \times \vec{v}| = \sqrt{6^2 + 6^2} = 6\sqrt{2}$$

$$A = \frac{1}{2} |\vec{u} \times \vec{v}| = 3\sqrt{2}$$

$$\boxed{\frac{1}{2} |\vec{u} \times \vec{v}|}$$

Geometric Properties of Determinants (Outline without proofs)

Consider the following figure:



$$\begin{aligned}
 \text{Area of } \square &= (a+c)(b+d) - 2\left(\frac{1}{2}ab\right) - 2\left(\frac{1}{2}cd\right) - 2(cb) \\
 &= ab + ad + bc + cd - ab - cd - 2cb \\
 &= ab - cb = \begin{vmatrix} a & b \\ c & d \end{vmatrix}
 \end{aligned}$$

Write $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} := \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

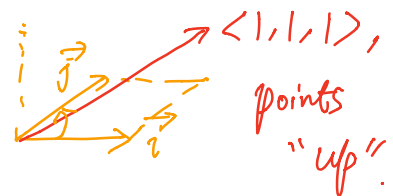
- $\left| \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right|$ is the area of the parallelogram spanned by $\langle a, b \rangle$ and $\langle c, d \rangle$. (Abs. values) (□)

- $\left| \det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & k \end{bmatrix} \right|$ is the volume of the parallelepiped spanned

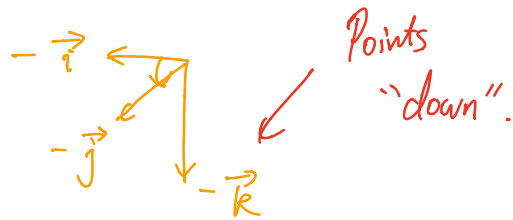
by $\vec{v}_1 := \langle a, b, c \rangle$, $\vec{v}_2 := \langle d, e, f \rangle$ and $\vec{v}_3 := \langle g, h, k \rangle$.

- For $\vec{v}_1, \vec{v}_2, \vec{v}_3$ in $\mathbb{R}^3$, $\det \begin{bmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vec{v}_3 \end{bmatrix} > 0$ if and only if $\vec{v}_1, \vec{v}_2$ and $\vec{v}_3$ the triple $(\vec{v}_1, \vec{v}_2, \vec{v}_3)$ satisfies the right-hand rule.

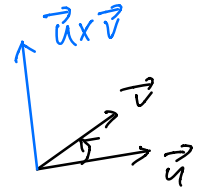
e.g. $\det \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1 > 0$,



$$\det \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = -1 < 0,$$



One question we have not addressed yet is : why $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ must satisfies the right-hand rule?
Suppose not parallel



Ans (Outline) : You can verify that

$$\det \begin{bmatrix} \vec{u} \\ \vec{v} \\ \vec{u} \times \vec{v} \end{bmatrix} = |\vec{u} \times \vec{v}|^2 > 0.$$

Properties of Cross Products

Can be proven algebraically

Properties of the Cross Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and r , s are scalars, then

- | | |
|--|---|
| 1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$ | 2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$ |
| 3. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$ | 4. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$ |
| 5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$ | 6. $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ |

It may be handy to memorize the following special cases:

$$\mathbf{i} \times \mathbf{j} = -(\mathbf{j} \times \mathbf{i}) = \mathbf{k}$$

$$\mathbf{j} \times \mathbf{k} = -(\mathbf{k} \times \mathbf{j}) = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = -(\mathbf{i} \times \mathbf{k}) = \mathbf{j}$$

$$\vec{u} \times \vec{v} \neq \vec{v} \times \vec{u}$$

Note that “ $\times$ ” is not commutative. It is also not associative:
 $(\mathbf{i} \times \mathbf{i}) \times \mathbf{j} \neq \mathbf{i} \times (\mathbf{i} \times \mathbf{j})$.

$$\begin{aligned} (a\vec{v} + b\vec{w}) \cdot \vec{u} &= 0 \\ a(\vec{v} \cdot \vec{u}) + b(\vec{w} \cdot \vec{u}) &= 0 \end{aligned}$$

"A geometric proof of $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$ ".

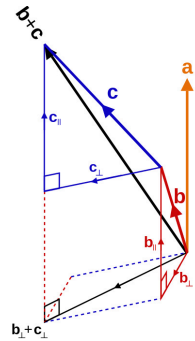


Diagram:
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